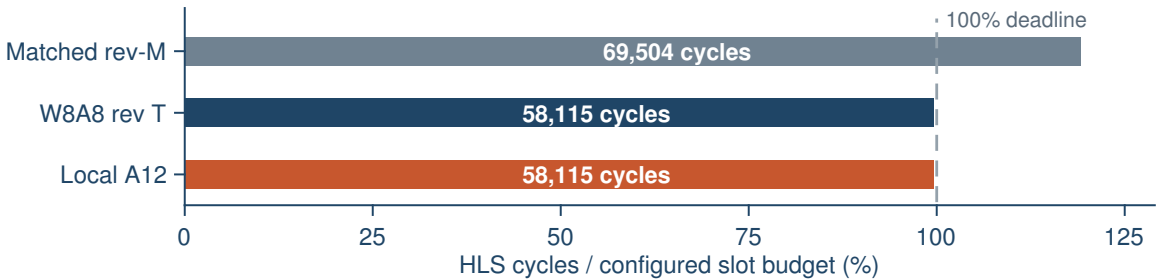
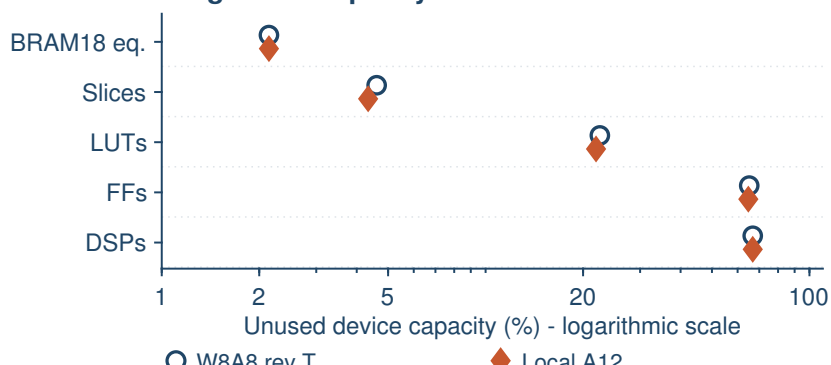
(a) Core HLS latency vs. slot budget



Matched rev-M / rev T: corrected-PDP checkpoint; HLS 2022.2; deadline: 58,333 cycles.

(b) Post-route remaining FPGA capacity



A12 remaining (raw units)
6
577
11,667
68,904
147

Local A12 - rev T: +320 LUT; +436 FF; +36 slices; +0 DSP; +0 BRAM18 eq.

Post-route WNS at 100 MHz: rev T +0.361 ns; A12 +0.115 ns.

A12 JTAG: finite-vector self-test passed; streaming and measured board power remain unverified.